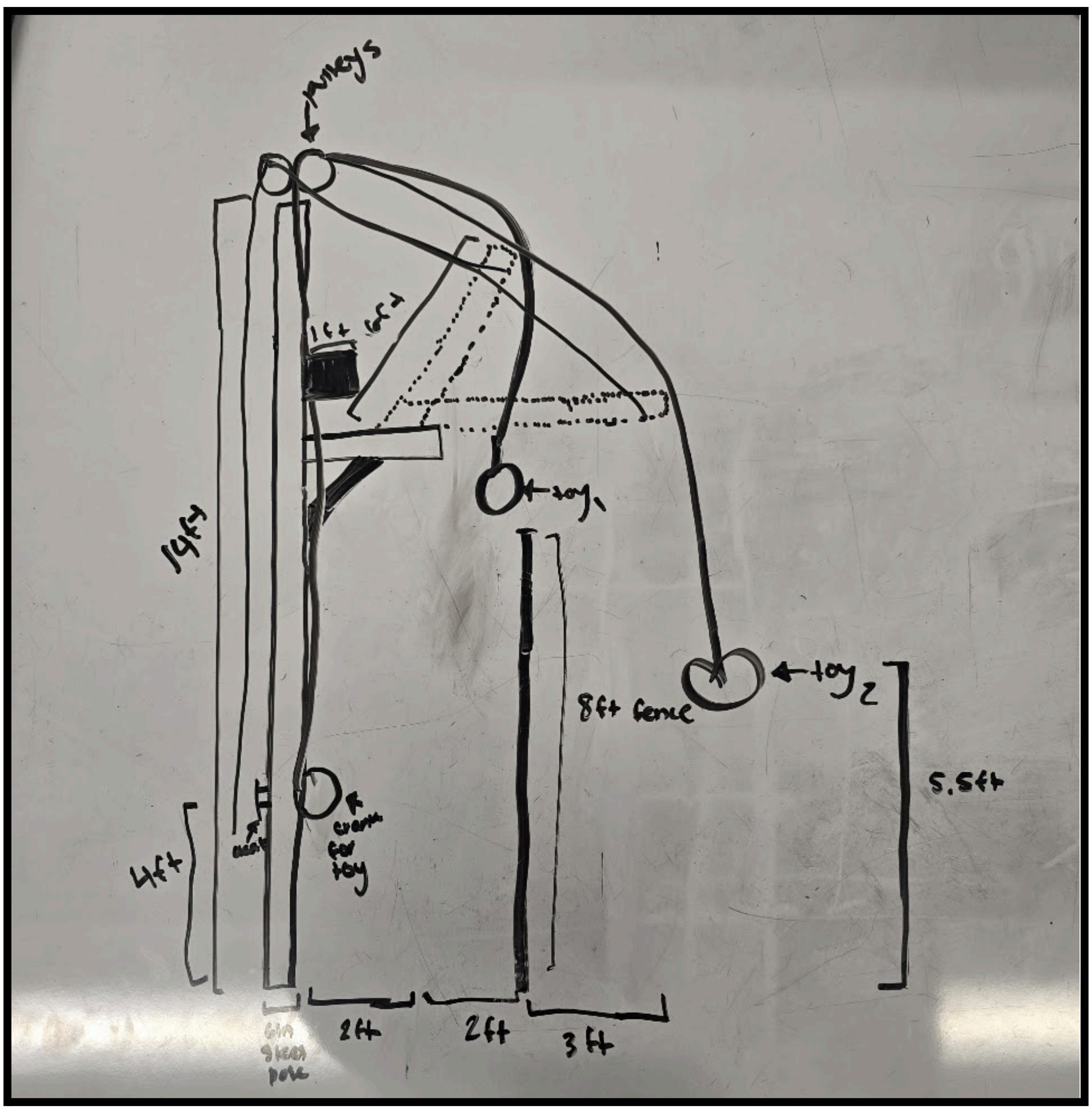
*Note: all prototype drawings are drawn with scales aligning with the above measurements to ensure their utility.

Individual Member Ideas:

We had each member individually decide on their best idea to present to the group for group decision making. Listed below are each group members' self-rated best ideas:

Student A:

Image:



Explanation:

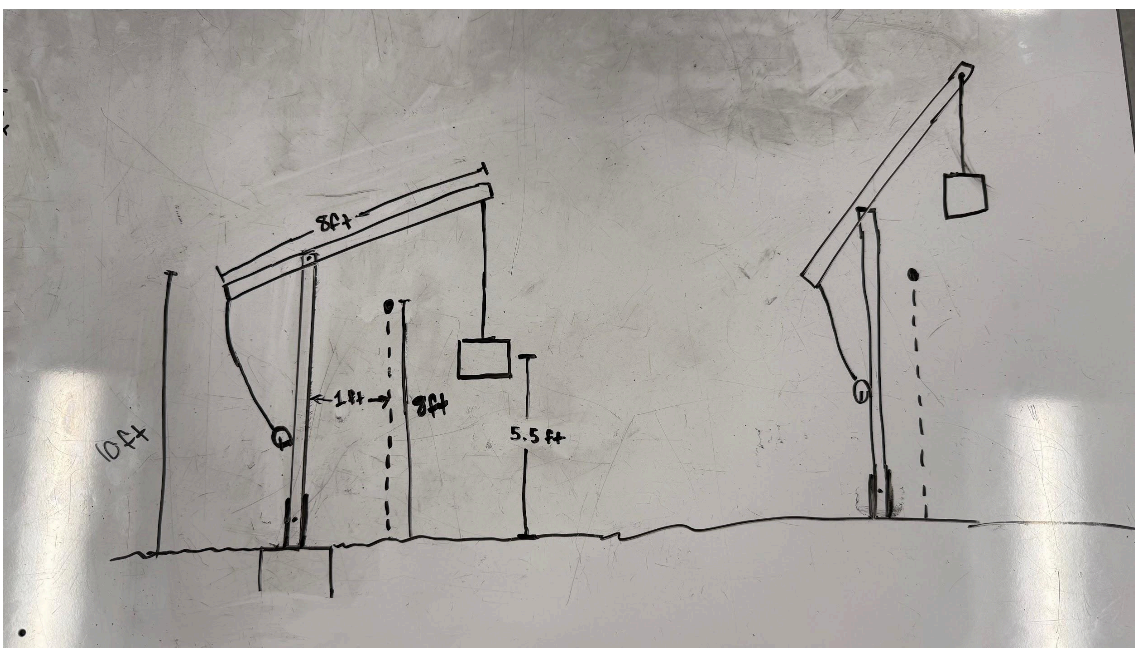
This design contains a block that will move on a pulley, where when it is in the upright position, it will allow the toy to cross to the human side of the fence, and when it goes down, it crosses to the animal side of the fence. When the toy is on the desired side, there is a separate pulley system for

Why:

I chose this design because it minimizes the moving parts necessary for our toy to make it across the fence. Instead of putting it on a rotating disk, this is also much more secure, as full rotation as opposed to a switch is generally deemed as stronger.

Student B:

Image:



Explanation:

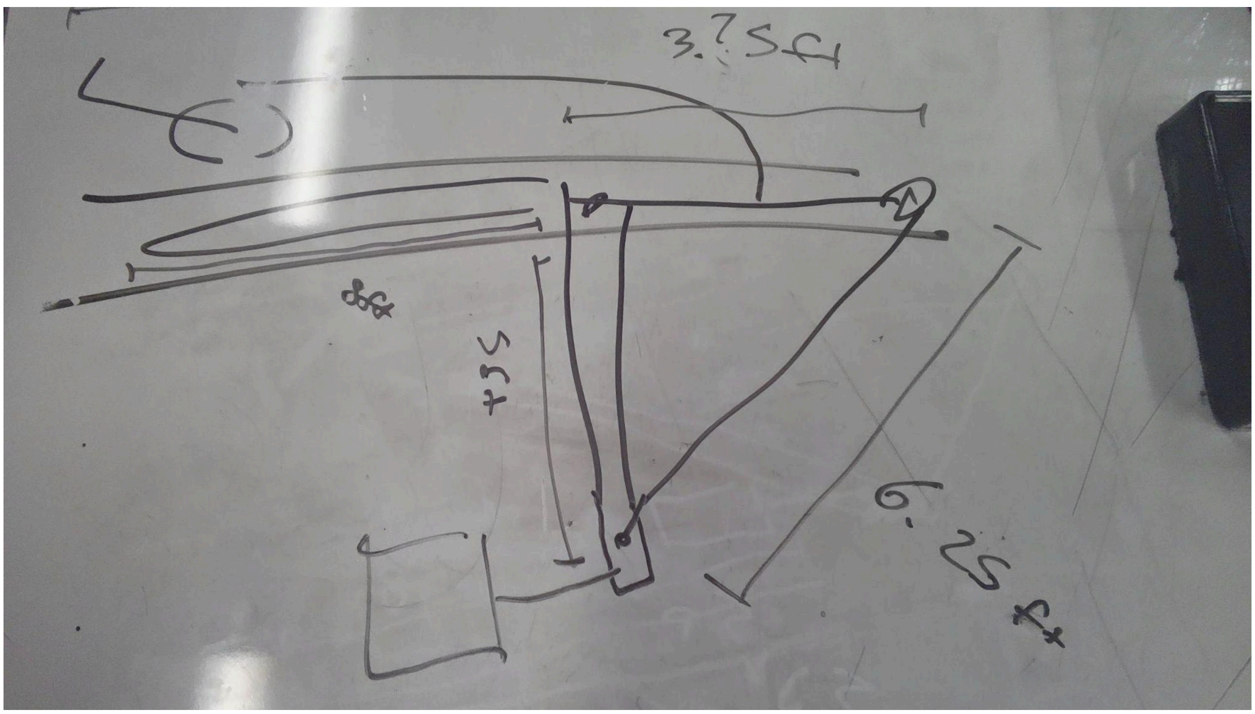
The device is constructed with two poles. The toy is suspended from a carabiner attached to the end of the upper pole that will go over the fence. A cable is attached to a crank mounted on the vertical pole and the outer end of the top pole. The top pole pivots at a point 3.5 feet from the outer end. The base is inserted into a pipe with a slightly larger diameter and a hole is drilled through both. When the toy is in place, a rod is inserted through the hole to prevent the device from rotating. When the crank is retracted, the toy end of the pole raises so it can clear the fence. The toy is then lowered by letting out more cable/rope.

Why:

This design limits moving parts and components that might break. It is made of two poles, a pivot of some kind, and a crank. The toy is moved over the fence by rotating the whole device instead of an additional rotating disk which provides more structural integrity.

Student C:

Image:



Explanation:

The device is composed of two poles and a string, with the string being connected to the tip and base of the horizontally extending pole, so that using the crank to move the string vertically will move the tip of the pole, and therefore the string, up and to the left. The string is connected to the base of the horizontally extending pole by going through the top of the vertical pole and connecting to the base of the horizontal pole by extending down the inside of the vertical pole. The crank is connected to the string by connecting to a string which then connects to the string that connects to the pole like the handle of a lasso is connected to the hoop of said lasso. It includes multiple cuts into the vertical pole for the horizontal pole to slide down it, and for the strings to go through the pole. The toy hangs from the end of the horizontal pole.

Why:

This design aims to simplify the mechanism into just one moving part, but in turn increases the amount of cuts into the metal pole and requires an additional stabilizing wire in the ground to stop the vertical metal pole from bending.

Decision Matrix (based on criterion from part c)

We ranked our projects on a scale of 1-3 since we were comparing 3 projects, with the greatest added total being the design that best met our requirements. We had a decision matrix for high priority and low priority criterion with high priority weighted at 1 and low priority weighted at 0.5. This was to ensure that our most important priorities are met first.

High Priority Matrix (weighted at 1)

Projects	Efficacy	Structural Integrity	Ability to Change Toys	Durability	Low Price	Totals
A	3	3	3	3	2	15
B	2	2	2	2	3	11
C	1	1	1	1	1	5

Low Priority Matrix (weighted at 0.5)

Projects	Ease of Maintenance	Aesthetics	Total:
A	2	2	4
B	1	3	4
C	3	1	4

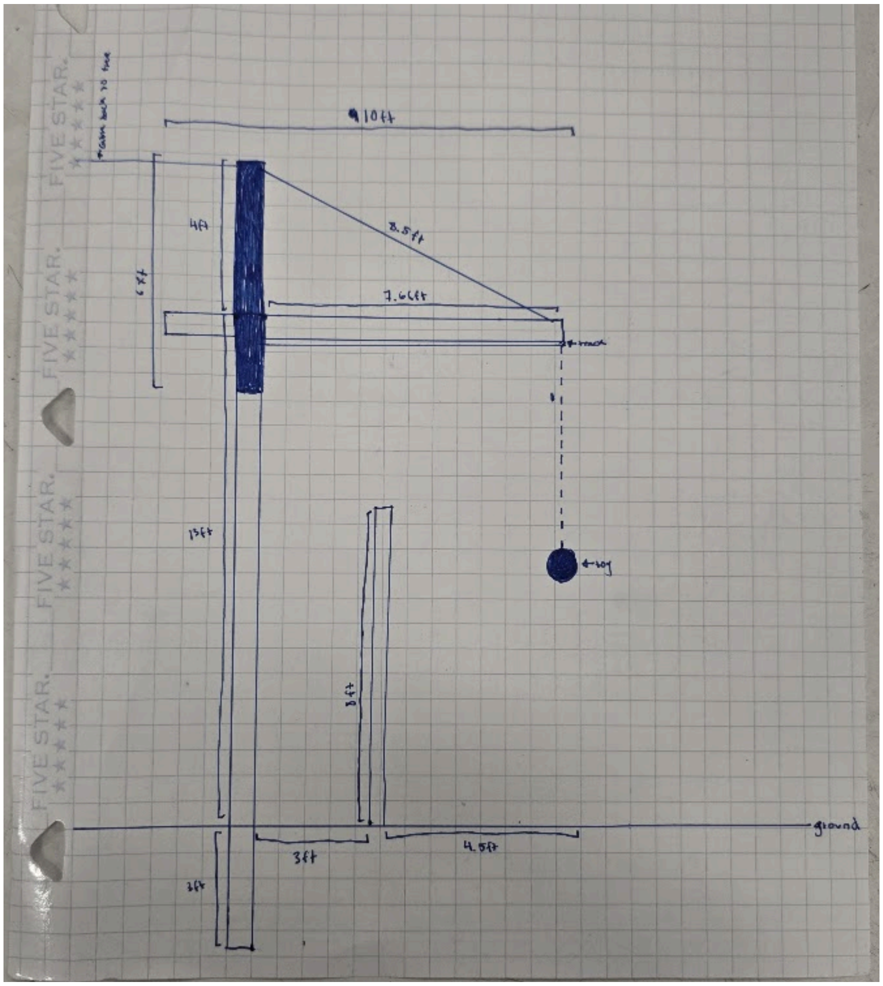
Calculated Total:

A = 15(1)+ 4(0.5) = 17
B = 11(1) + 4(0.5) = 13
C = 5(1) + 4(0.5) = 7

Thus, our chosen design is Student A's.

Final Design Decision:

Image:



Explanation:

Our final design consists of a 4x4x16 cemented 3 feet into the ground positioned 3 feet away from the 8 foot fence. This system has a cantilever 13 feet above the ground extending out 7.66 feet which has a track attached beneath it helping to regulate our pulley system. To hold this cantilever in place (from moving side to side), we added 2 2x4x6's on the side of our device and one cut in the middle placed with 8 inch bolts. Then, a cable is attached from the top side of these 4x4's and attached to the end of the cantilever. There is also a support cable going back from this top point. For the male enclosure, this goes back to a tree, and for the female enclosure, this goes back to the ground. The pulley system is a dual pulley system, with one on a hand crank that raises and lowers the toy and one on a simple hand pulley where you pull the toy from side to side.

Why:

We wanted our design to be able to move the toy from side to side and up and down and a pulley system which was already partially in place was the best way to do so. Additionally, we wanted to support the existing structure for cost reasons, and that is why the cables were necessary.

Plan of Action:

Task:	3/16-20	3/23-27	3/30-4/3	4/6-10	4/13-17	4/20-14	4/27-5/11
CAD and Sketch							
Element E							
Element F							
Acquire Materials							
Build Hanger							
Element G							
Element H							
Element I							
Element J							
Element K							
Element L							

Specifics:

Element E:

1. Describe in detail how the proposed solution is supported by STEM concepts including core disciplinary ideas and mathematical practices.
2. Describe the STEM concepts as they apply to the proposed design and explain how they meet design requirements (for at least 4 criterion).
3. Include any relevant calculations.
4. Use these stem practices to justify design decisions.
5. Obtain expert review from 2-3 experts of the STEM connections (cannot be a stakeholder). Write their input in the document, what changes they would make, and how the design will reflect those changes.
6. Obtain a field expert signature and explain what qualifies them as an expert.

Element F:

1. Provide evidence to show that design in viable for each criterion.
2. Do a risk assessment and mitigation plan for each component of the system and how that may impact meeting the criterion.

Acquire Materials:

From Homedepot's website:

1. $25.08 - 6 2x4x8 - https://www.homedepot.com/p/2-in-x-4-in-x-8-ft-2-Ground-Contact-Pressure-Treated-Southern-Yellow-Pine-Lumber-106147/206970948
2. $6.28 - 8 flat rounded head stainless steel bolts 4 inches in length - https://www.homedepot.com/p/Everbilt-1-4-in-20-x-4-in-Zinc-Plated-Phillips-Flat-Head-Machine-Screw-2-Pack-831621/317479108
3. $33.22 - 10 8 inch stainless steel bolts - https://www.homedepot.com/p/Everbilt-3-8-in-x-8-in-Stainless-Steel-304-Hex-Bolt-5-Pack-868570/321055761
4. 18 washers (10 large and 8 small) - Mr. Kiser's classroom
5. $50.70 - 6 tensioner cables for the lines coming down - https://www.homedepot.com/p/Everbilt-1-4-in-x-5-1-4-in-Stainless-Steel-Hook-and-Eye-Turnbuckle-803724/205883056
6. $9.70 - 2 bags concrete https://www.homedepot.com/p/Quikrete-60-lb-Concrete-Mix-110160/100318478
7. $55.16 -100 feet 1/16th inch coated galvanized steel cable (4 for pulley systems) https://www.homedepot.com/p/Everbilt-1-16-in-x-50-ft-Galvanized-Vinyl-Coated-Steel-Wire-Rope-811062/300019543

8. $47.92 - 100 feet 1/8th inch galvanized steel cable (Male enclosure to tree = 255 inches. Female enclosure to tree ~ 110 inches. Each cantilever support to the cantilever = 732 inches) https://www.homedepot.com/p/Everbilt-1-8-in-x-50-ft-Galvanized-Uncoated-Steel-Wire-Rope-803152/203958867
9. 10 cable crimps (1/16ths inch) - Mr. Kiser's classroom
10. $15.88 - 8 Cable Crimps - 1/8th of an inch https://www.homedepot.com/p/Everbilt-1-8-in-Aluminum-Ferrule-and-Stop-Set-43254/205887928
11. $45.96 - 12 eyebolts - https://www.homedepot.com/p/Everbilt-1-4-in-x-2-in-Stainless-Steel-Eye-Bolt-with-Nut-2-Pack-803554/205883084
12. Cable cutter and crimper - Joseph, Marten, and Mr. Kiser's classroom
13. $35 - delivery fee

Total: $324.90

Build Toy Hanger:

1. Purchase Materials
2. Drill the 3 2x4's together in the way they are oriented in the sketch/CAD.
3. Drill an eyebolt into the back of this apparatus and the sides (for the cables going to the tree and the cables going down onto the cantilever.
4. Drill eyebolts into the sides of the cantilever.
5. Drill the 2x4's through the vertical with 8 inch bolts.
6. Secure the track with the 4 inch bolts.
7. Secure the hanger by attaching the 1/8th inch galvanized steel cables to the tree and to itself (use the cable cutter, crimper, crimps, and the tensioner cables).
8. Rig the pulley systems with the 1/16th inch coated galvanized steel cable.
9. Do this again for the female enclosure, but secure the back cable to the ground with an eyebolt in cement as opposed to the tree.

Element G:

Take notes or pictures of the full construction process (all images require captions)

Record details about all steps of construction, including the tools and manufacturing methods used.

Take note of the progression from blueprint to various physical prototypes including why and how aspects of your design have changed.

Present the details of your final prototype (list of material details (item name, description, where it was purchased, cost per unit, number of units, total cost)), detailed sketches and CAD).

Ensure the prototype is sufficiently functional for testing.

Obtain field expert review from 2-3 experts for all aspects of the design that cannot be physically or mathematically modeled, and their approval of this notion.

Element H:

1. Create a testing plan for the prototype against the high and low priority design criterion. (explain step by step plan for testing following scientific best practices and 7 trials per test (show how the test is set up, what happens during the test, the expected range of results and error, the minimum result that would meet the criterion, and goals past the criterion).
2. Obtain field expert review from 2-3 experts and notate the changes they would make.
3. Recreate your testing plan with the expert review in mind.

Element I:

1. Restate high and low priority criteria and justify each test in the testing plan.
2. Conduct tests and record results.
3. Connect the results with the success or failure of your design with appropriate charts, graphics, spreadsheets, etc.
4. Explain the implications of the collected data.
5. Construct your next steps for the design process.

Element J:

1. Document feedback of field experts and stakeholders.
2. Provide clear prompts to stakeholders and field experts for this feedback.
3. Summarize evaluators comments for trends and implications.

Element K:

1. Summarize each step of the design process (all elements).
2. Explain what went well and what didn't for each project step and why.
3. Write your lessons learned from this experience. Write what future teams or engineers could do better.
4. Write how the design could be improved for the future.

Element L:

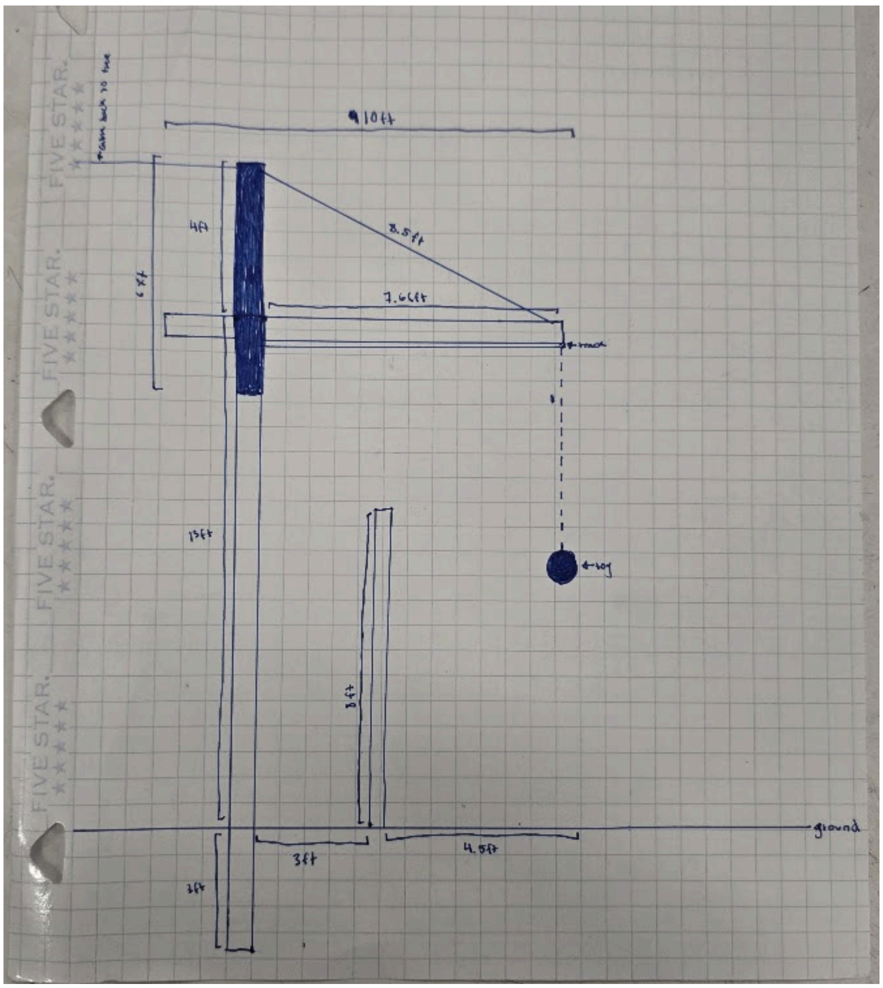
1. Describe recommendations for further developments if the project were to continue.
2. Explain what aspects of the project you would change versus keep the same, what would have made the project easier, what designs you have made.
3. Explain how these recommendations could be implemented in the future.

Blueprint:

CAD link:

https://cad.onshape.com/documents/3f842061d2db72545317d873/w/3c34cad911991e55a830b20b/e/b8a9c19b4cb6c5bba7d710bd?renderMode=0&uiState=69bb21cd4c33f74d850dc045

Sketches (with dimensions):



Manufacturing Methods:

- Circular saw to cut 2x4's to the correct lengths
- Drill to drill the bolts and eyebolts through the wood
- Tensioner cables to tighten the cables to the correct length
- Cable Cutters and Crimpers to secure the cables
- Ladder to reach the apparatus

Material Choices:

- We chose wood, steel bolts, and galvanized steel cable.

Costs:

From Homedepot's website:

14. $25.08 - 6 2x4x8 - https://www.homedepot.com/p/2-in-x-4-in-x-8-ft-2-Ground-Contact-Pressure-Treated-Southern-Yellow-Pine-Lumber-106147/206970948
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